

Amal Omrani

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Profile

PhD candidate in financial mathematics at Université Paris Dauphine (CEREMADE). My research focuses on dynamic programming methods for computing hedging prices under transaction costs.

Research

PhD Thesis

2023–Present

Université Paris Dauphine, CEREMADE

Dynamic programming and numerical computation of superhedging prices under proportional transaction costs. Supervisor: Prof. Emmanuel Lepinette.

Research Master

2023

Master Projects – LIMOS, Université Clermont Auvergne

Clustering approaches based on reduced Gaussian processes for regression and classification (focus on reducing computational and memory complexity).

GitHub

Supervised by: Prof. Chafik Samir Co-supervised by: Anis Fradi

Publications

2025

Lepinette, E., & Omrani, A. Explicit Recursive Construction of Super-Replication Prices under Proportional Transaction Costs, arXiv:2503.02419

2024

Omrani, A., Fradi, A., & Samir, C. Reduced Run-time and Memory Complexity Regression with a Gaussian Process Prior, In JDS

2023

Ben Amor, M. A., & Omrani, A. Chernoff's inequality in Riesz spaces, Quaestiones Mathematicae

Seminars and Conferences

SMFP-2025, SCF-2025, WQF-2025, Bachelier-2025, JDS-2024, CIMPA-2024, RGOSA-2023

Teaching

2025–2026

Introduction to Finance, Université Paris Dauphine – Financial markets and derivatives, European options (30 students, 40h)

2024–2025

Statistics L2, Université Paris Dauphine – Probability, sampling and estimation (30 students, 50h)

2024–2025

AI Projects, Université Paris Dauphine – Data analysis (Python) (30 students, 15h)

2019–2023

Algebra I and II, Preparatory Institute for Engineering Studies, Monastir (60 students/year, 18h/week)

Education

2023–Present

PhD in Financial Mathematics, Université Paris Dauphine, CEREMADE

Dynamic programming and implementation of financial models from economics and finance

2022–2023

Master in Mathematics and Applications, Université de Monastir, Tunisia (Rank 1)

Thesis and internship at LIMOS Laboratory, Université Clermont Auvergne

2019

Mathematics Agrégation — Probability and Statistics, ENS Tunis

Skills

Technical Python, C++, MATLAB, $\LaTeX$

Soft skills Scientific communication, mentoring, teamwork, organization

Organization Co-organizer of Young Researchers' Days at CEREMADE (YRSD 2024–2025)

Languages French (fluent), English (fluent), Arabic (fluent), Italian (basic)